

Intent-Driven Manifold Constraint: A Probabilistic Framework for Vision-Based Precision Assembly without Force Sensors

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Abstract—High-quality demonstration data is the cornerstone of general-purpose robot learning. However, current teleoperation paradigms present a dilemma: high-fidelity master-slave systems (e.g., ALOHA) ensure precision but entail high hardware costs, while low-cost vision-based retargeting methods (e.g., AnyTeleop) democratize data collection but suffer from low precision and “contextual blindness” in tight-tolerance tasks. To bridge this gap, we propose VIST (Vision-based Intent-aware STate estimation), a probabilistic framework that enables precision assembly using low-cost cameras and anthropomorphic arms without force sensors. Central to VIST is a continuous *intent factor* α , which real-time infers the semantic trend of the system state converging from 6-DoF free-space motion to a low-dimensional constrained manifold. This factor dynamically drives the covariance scheduling of an Adaptive Kalman Filter (AKF). Through this mechanism, VIST mathematically unifies *physiological tremor rejection*, *virtual manifold adsorption*, and *geometric subspace constraints* within a single estimator. Experiments on a sub-millimeter USB insertion task demonstrate that VIST significantly improves the success rate of low-cost teleoperation. Furthermore, comparative analysis verifies that VIST acts as a physical data regularizer, generating high-consistency datasets that significantly enhance the sample efficiency and robustness of downstream Imitation Learning policies (e.g., Diffusion Policy).

I. INTRODUCTION

Robot learning from demonstration (LfD) has witnessed a paradigm shift from early behavioral cloning [39], [40] to modern generative policies that model multi-modal distributions [?], [33], [41]. High-quality demonstration data is widely recognized as the critical bottleneck for scaling these general-purpose capabilities [35]. To address this, high-end master-slave systems (e.g., ALOHA [33], [49]) have achieved impressive results in precision manipulation by leveraging hardware isomorphism and force feedback. However, their prohibitive cost and rigid structural setups limit widespread adoption. Conversely, low-cost vision-based retargeting approaches, such as AnyTeleop [?], UMI [45], and DexPilot [50], have democratized data collection using consumer-grade cameras. Yet, these systems fundamentally lack the “contextual awareness” required for tight-tolerance tasks (e.g., sub-millimeter assembly), often reducing teleoperation to a frustrating trial-and-error process dominated by human physiological tremors.

We argue that the failure of low-cost systems in precision tasks is not merely a hardware deficiency, but a *cognitive*

misalignment. Unlike high-precision industrial robots that rely on absolute positioning, human-in-the-loop systems must compensate for sensing uncertainties through intelligent shared control [1]. Specifically, precision assembly is not a uniform motion process, but a dynamic transfer of *belief confidence* on the Lie Group $SE(3)$ [14]. This process involves two distinct semantic phases:

- 1) **Free-Space Exploration:** The human operator possesses maximum global planning confidence. The system should remain transparent, allowing 6-DoF exploration.
- 2) **Manifold Convergence:** As the end-effector approaches the target, environmental geometric constraints (e.g., the insertion axis) gain high local confidence. The system should actively intervene, compressing the search space from 6-DoF to a lower-dimensional sub-manifold (e.g., 1-DoF translation) to compensate for human jitter [15].

Existing teleoperation frameworks, however, fail to effectively manage this “**Semantic Duality**”—the contradiction between discrete task phases and the need for continuous physical interaction. This failure manifests in three critical deficits:

1. Contextual Blindness and Cognitive Overload. Current vision-based interfaces are often “geometry-agnostic” [5]. They apply uniform motion mapping gains regardless of whether the robot is moving freely or performing sub-millimeter alignment. For instance, in the final 5mm of a USB insertion, physical laws dictate that motion should be constrained to the Z-axis. Yet, existing systems blindly permit large lateral deviations caused by hand tremors. This forces the operator to expend excessive cognitive effort manually suppressing physiological noise to fight the system’s omnidirectional gain, leading to performance degradation predicted by Fitts’ Law [6], [7].

2. Dynamic Discontinuity in Discrete Handoff. To address the need for constraints, traditional shared control methods often employ Finite State Machines (FSM) or distance-based thresholds to switch between “free” and “assisted” modes [2], [11]. However, such discrete switching induces instantaneous mutations in force or position (Jerk) at the boundary of the capture region. This “hard handoff” creates a jarring sensation—described as hitting a virtual wall—which disrupts the operator’s sensorimotor loop and introduces unnatural kinematic artifacts into the recorded trajectory [10], [12].

3. The Decoupling of Subjective Feel and Data Quality. Perhaps the most overlooked issue is the conflation of

*This work was supported by University Research Grants.

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“natural teleoperation feel” with “high-quality training data.” Analytical retargeting algorithms, such as the “4+3” **Swivel-Angle Mapping** used in our baseline [51], effectively avoid the singularities of numerical IK and produce human-like arm configurations. However, this “open-loop” biomimetic mapping is a double-edged sword: it faithfully preserves kinematic feasibility but also faithfully records *nuisance stochasticity* (e.g., tremors, hesitation) in task-critical dimensions. For generative models like Diffusion Policy [?], this high-entropy distribution near the task manifold creates **Causal Confusion** [36]. The model risks learning the operator’s noise as a complex “spiral search strategy,” leading to jittery or unsafe behaviors during autonomous execution.

To bridge these gaps, we propose VIST (**V**ision-based **I**ntent-aware **S**Tate estimation). Instead of treating intent as a discrete switch, we model it as a continuous **Intent Factor** $\alpha \in [0, 1]$, which quantifies the semantic trend of the system state converging from high-dimensional free space to a constrained low-dimensional manifold. This factor drives an Adaptive Kalman Filter (AKF) [18] to dynamically schedule the covariance matrices, mathematically unifying signal denoising, virtual guidance, and manifold constraints without external force sensors [17], [20].

The main contributions of this work are three-fold:

- **Semantic-Geometric Intent Modeling:** We propose a continuous intent inference model that fuses geometric distance, kinetic energy, and vector alignment. It serves as a semantic interpreter, translating vague human motions into a confidence metric for manifold convergence, effectively resolving contextual blindness [3].
- **Continuous Covariance Scheduling:** We introduce an α -driven adaptive mechanism for the Kalman Filter process (Q) and observation (R) noise. This establishes a mapping from high-level semantics to low-level control parameters, converting “hard switching” into a probabilistic “soft landing,” ensuring C^1 continuity in control authority.
- **Physical Data Regularization:** We demonstrate that VIST acts as a physical filter for data collection. By projecting nuisance variance into the null space of the task manifold [15], it generates low-entropy, high-consistency datasets. Experiments show that policies trained on VIST-collected data significantly outperform baselines in sample efficiency and robustness for precision assembly tasks.

II. RELATED WORK

A. Vision-Based Teleoperation and Geometric Semantics

Recent advancements in vision-based teleoperation have democratized data collection by replacing expensive haptic devices with low-cost cameras [?], [45], [50]. Systems like AnyTeleop [?] and UMI [45] utilize kinematic retargeting to map human hand poses to robot joint spaces effectively. However, these frameworks often operate under a “contextual blindness”: they apply uniform motion mapping gains regardless of whether the operator is moving freely or attempting

precision alignment. In precision assembly, the task space undergoes a dramatic geometric transition from free space ($SE(3)$) to a low-dimensional manifold (e.g., a line for peg-in-hole). Current systems lack the ability to perceive these *geometric semantics* [5]. Consequently, operators are forced to manually suppress physiological tremors to counteract isotropic gains, leading to “cognitive overload” as described by Fitts’ Law [6], [7]. Our work addresses this by introducing the Intent Factor α , serving as a semantic bridge that modulates control parameters based on the alignment between human intent and environmental constraints.

B. Shared Control and Continuous Arbitration

To assist in precision tasks, shared control frameworks typically arbitrate authority between the human and the robot [1]. Traditional methods often rely on Finite State Machines (FSM) or distance-based thresholds to switch between “free” and “assisted” modes [2], [11]. However, such discrete switching introduces dynamic discontinuities (jerk) at the handoff boundary, creating haptic conflicts or “deadlocks” [10]. While Adaptive Kalman Filters (AKF) are widely used for state estimation [18], they predominantly rely on signal residuals (innovation) to tune covariance. This reactive approach is insufficient for proactively shaping control authority. VIST proposes a novel formulation where the high-level intent factor α directly drives the inverse covariance scheduling of the process noise Q and measurement noise R . This mathematically transforms the “hard switching” of FSMs into a probabilistic “soft landing” [12], ensuring C^1 continuity in the control loop even without force sensors [17], [20].

C. Data Quality and Policy Learning

A critical yet overlooked discrepancy exists between subjective teleoperation playability and objective data quality. State-of-the-art retargeting algorithms, such as the “4+3” analytical decoupling used in [51], effectively resolve kinematic singularities and produce visually natural motions. However, this “open-loop kinematic feasibility” is a double-edged sword: it faithfully records the operator’s nuisance stochasticity (e.g., hand tremors) during constrained task phases. For generative imitation learning models like Diffusion Policy [?] or ACT [33], high-entropy distributions near the task manifold can induce *causal confusion* [36]. The model may spuriously correlate physiological jitter with task success, leading to ambiguous policies. As highlighted by Mandlekar et al. [35], data quality often outweighs quantity. VIST acts as a physical regularizer that performs “semantic filtering” at the signal source. By projecting the task manifold constraints into the null space of the Jacobian ($Q = J^\dagger \Sigma J^{\dagger\top}$) [15], VIST compresses invalid multimodal noise while preserving essential approach diversity, thereby providing downstream models with unambiguous geometric signals.

III. METHODOLOGY

A. Problem Formulation and Manifold Geometry

We formulate the precision assembly task as a stochastic state estimation problem on the Lie Group $SE(3)$. Let the

robot end-effector state at time k be $\mathbf{T}_k \in SE(3)$, comprising rotation $\mathbf{R}_k \in SO(3)$ and translation $\mathbf{p}_k \in \mathbb{R}^3$. The objective is to guide the system from a high-dimensional free space manifold $\mathcal{M}_{free} \cong SE(3)$ to a task-constrained sub-manifold $\mathcal{M}_{task} \subset SE(3)$ of dimension $d < 6$ (e.g., a 1-DoF line for peg-in-hole) [14].

1) *Observation Model and Stochastic Tremor*: Unlike industrial automation, we assume the reference trajectory is provided by a human operator via a vision-based interface. While kinematic retargeting algorithms (e.g., "4+3" mapping [51]) ensure geometric feasibility, they operate as open-loop observers. Thus, the operator's input $\mathbf{z}_{human}$ inherently carries physiological noise $\mathbf{w}_{tremor}$, which becomes detrimental near the task manifold:

$$\mathbf{z}_{human} = h(\mathbf{x}_{human}) + \mathbf{w}_{tremor}, \quad \mathbf{w}_{tremor} \sim \mathcal{N}(0, \Sigma_{tremor}(\mathbf{x})) \quad (1)$$

Crucially, Σ_{tremor} is state-dependent: as noted by Fitts' Law [6], [7], the signal-to-noise ratio (SNR) degrades significantly as the task difficulty (precision requirement) increases.

B. Semantic-Geometric Intent Modeling

A core challenge in shared control is the "semantic gap": raw motion does not explicitly convey whether the operator intends to *explore* or *insert*. To bridge this, we introduce a continuous **Intent Factor** $\alpha_k \in [0, 1]$. Instead of heuristic thresholding, we derive α_k based on the **Principle of Least Action** in the phase space $(\xi, \dot{\xi})$, decomposing intent into three physically grounded components:

1) *Geometric Potential (Spatial Consistency)*: We view the task manifold $\mathcal{M}_{task}$ as the global minimum of a virtual potential field. The geometric intent α_{geo} measures the probabilistic depth of the current state within this field. Defining the error twist $\xi_{err} = \ln(\mathbf{T}_{target}^{-1} \mathbf{T}_{curr})^V \in \mathfrak{se}(3)$ [14], we formulate:

$$\alpha_{geo} = \exp\left(-\frac{1}{2} \xi_{err}^\top \mathbf{W}_{task} \xi_{err}\right) \quad (2)$$

where $\mathbf{W}_{task}$ is the metric tensor defined by physical assembly tolerances. This term ensures the system only activates assistance when spatially aligned [5].

2) *Kinematic Energy (Temporal Stability)*: Precision operations fundamentally require low kinetic energy. High-velocity motions indicate exploration or transport, corresponding to high entropy. We define the velocity intent α_{vel} using a soft-switching function of the twist norm $\|\xi_{vel}\|$:

$$\alpha_{vel} = \frac{1}{1 + \beta_v \|\xi_{vel}\|_{\mathbf{M}}} \quad (3)$$

This term provides **Pseudo-Haptic feedback**: upon rigid contact (collision), physical constraints force $\|\xi_{vel}\| \rightarrow 0$, driving $\alpha_{vel} \rightarrow 1$. As we will show in Sec. III-C, this triggers the filter to increase stiffness, mathematically mimicking force feedback without torque sensors [17].

3) *Vector Field Alignment (Active Gating)*: To resolve conflicts (e.g., intentional withdrawal near the hole), we introduce a directional gating term α_{dir} based on the cosine similarity between the velocity vector field and the geodesic error gradient:

$$\alpha_{dir} = \frac{1}{2} \left(1 + \frac{\xi_{vel}^\top \xi_{err}}{\|\xi_{vel}\| \cdot \|\xi_{err}\|} \right) \quad (4)$$

This acts as a "veto mechanism" [1]: if the operator moves away from the target ($\cos \theta < 0$), α_{dir} suppresses the overall intent, preventing the robot from fighting the human.

The aggregated Intent Factor is fused as a gated product:

$$\alpha_k = \underbrace{\sigma(w_g \alpha_{geo} + w_v \alpha_{vel})}_{\text{State Prior}} \cdot \underbrace{(\alpha_{dir})^\eta}_{\text{Active Gate}} \quad (5)$$

where $\eta \geq 1$ modulates the sensitivity of the directional veto.

C. Intent-Driven Manifold-Constrained Estimation

To address the "hard-switching" discontinuity in traditional FSM-based shared control [2], [11], we propose VIST (Vision-based Intent-aware State Estimation). We model the system as a time-varying stochastic process where the Intent Factor α_k dynamically schedules the covariance structures of an Adaptive Kalman Filter (AKF).

1) *Process Model with Null-Space Projection*: We define the state $\mathbf{x}_k = [\mathbf{q}_k, \dot{\mathbf{q}}_k]^\top$ in the joint space to ensure kinematic feasibility. The prediction step follows a constant-velocity model with α -modulated uncertainty:

$$\mathbf{x}_{k|k-1} = \mathbf{F} \mathbf{x}_{k-1} + \mathbf{w}_k(\alpha), \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k(\alpha)) \quad (6)$$

Unlike standard filters, our process noise $\mathbf{Q}_k(\alpha)$ is not constant. We construct a task-space covariance Σ_{task} that smoothly interpolates between isotropic exploration (Σ_{free}) and low-rank manifold constraints (Σ_{cons}) based on α . To rigorously enforce these constraints on the robot's kinematics, we project Σ_{task} into the joint space using the Jacobian pseudo-inverse $\mathbf{J}^\dagger$:

$$\mathbf{Q}_k(\alpha) = \mathbf{J}^\dagger(\mathbf{q}) [(1 - \alpha) \Sigma_{free} + \alpha \Sigma_{cons}] \mathbf{J}^{\dagger\top}(\mathbf{q}) + \varepsilon \mathbf{I} \quad (7)$$

Theoretical Insight: As $\alpha \rightarrow 1$, Eq. 7 performs a "soft attention" mechanism [15]. It mathematically freezes the variance in the null space of the task manifold (e.g., prohibiting X/Y motion) while maintaining high variance along the task axis (e.g., Z insertion), effectively creating a *probabilistic virtual fixture* [22].

2) *Observation Model with Inverse Covariance Scheduling*: We treat the human input $\mathbf{z}_h$ and the virtual guidance $\mathbf{z}_v$ (derived from perception) as competing observations. The effective measurement covariance $\mathbf{R}_k(\alpha)$ is scheduled to arbitrate authority:

$$\mathbf{R}_h(\alpha) = R_{base} \cdot e^{\lambda \alpha} \cdot \mathbf{I} \quad (8)$$

$$\mathbf{R}_v(\alpha) = \frac{R_{min}}{\alpha + \varepsilon} \cdot \mathbf{I} + \gamma \|\mathbf{z}_h - \mathbf{z}_v\|^2 \cdot \mathbf{I} \quad (9)$$

Eq. 8 implies that as intent increases, the system "trusts" the noisy human input less (high $\mathbf{R}_h$), relying instead on

the manifold-constrained process model. Eq. 9 contains two competing terms: the first term creates an "adsorption force" by reducing guidance noise as $\alpha \rightarrow 1$; the second term is a "break-away" function that inflates uncertainty if the human significantly diverges from the guidance [3], ensuring safety and overriding capability.

3) *The Update Step as Variable Admittance Control*: The final state update $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_{syn} - \mathbf{H}\hat{\mathbf{x}}_{k|k-1})$ is governed by the Kalman Gain $\mathbf{K}_k(\alpha)$. Through this formulation, $\mathbf{K}_k$ acts as a *variable stiffness matrix*. In free space ($\alpha \approx 0$), high $\mathbf{Q}$ and low $\mathbf{R}_h$ yield high gain (transparency). On the manifold ($\alpha \approx 1$), the structured $\mathbf{Q}$ and low $\mathbf{R}_v$ force the gain to reject disturbances orthogonal to the task, achieving C^1 -continuous control authority transition without explicit state machine switching [18].

IV. EXPERIMENTS

We designed experiments to validate two core hypotheses:

- 1) **Teleoperation Efficiency**: VIST reduces operator cognitive load and physical effort in high-precision assembly tasks compared to kinematic-only baselines.
- 2) **Data Quality for Learning**: Demonstrations collected via VIST possess lower entropy and higher consistency, leading to faster convergence and higher success rates for downstream imitation learning policies (e.g., Diffusion Policy).

A. Experimental Setup

1) *Hardware System*: Our teleoperation testbed consists of a **Franka Emika Panda** robotic arm equipped with a parallel-jaw gripper. The operator hand motion is captured by a single **Intel RealSense D435i** depth camera mounted on a tripod. We utilize MediaPipe for hand tracking and apply the "4+3" swivel-angle analytical inverse kinematics [44] as the base retargeting algorithm. The system runs on a workstation with an Intel i9-13900K CPU and NVIDIA RTX 4090 GPU. The control loop operates at 60Hz.

2) *Task Design*: We selected three contact-rich precision tasks with increasing difficulty (tolerance $< 2\text{mm}$):

- **T1: USB Insertion**: Inserting a USB-A connector into a hub. Requires strict 1-DoF alignment and force modulation.
- **T2: 12-Pin Aviation Plug**: Aligning a circular connector with a specific keyway orientation. Requires coupled rotation-translation control.
- **T3: Gear Assembly**: Placing a gear onto a shaft with tight tolerance. Requires handling occlusion and collision.

3) *Baselines*: We compare VIST against three state-of-the-art teleoperation frameworks:

- **Direct Teleop (Raw)**: Direct mapping of hand pose to robot TCP using differential IK without filtering.
- **One Euro Filter [21]**: The standard signal processing baseline used in VR/AR, tuned for jitter reduction ($\beta = 0.05, f_{min} = 1.0$).

- **AnyTeleop [44]**: A strong SOTA baseline using optimization-based retargeting and collision avoidance, but lacking intent-aware dynamics.

B. Evaluation I: Teleoperation Performance

We recruited 10 participants (4 experts, 6 novices). Each subject performed 10 trials per task per method (Total $N = 1200$ trials).

1) Metrics:

- **Success Rate (SR)**: Percentage of successful insertions within 60 seconds.
- **Completion Time (CT)**: Time duration for successful trials.
- **Normalized Jerk (NJ)**: Defined as $\int |\ddot{\mathbf{x}}|^2 dt$, measuring the smoothness of the motion.
- **Contact Force (CF)**: Estimated interaction force at the end-effector (derived from joint torques). Lower is better for safety.

2) *Results Analysis*: Quantitative results are summarized in Table I. VIST significantly outperforms baselines in precision tasks. In the USB Insertion task (T1), VIST achieved a **95% success rate**, compared to 72% for AnyTeleop. Crucially, the **Normalized Jerk** metric decreased by **64%** compared to the One Euro Filter. This supports our claim that VIST's covariance scheduling effectively filters out physiological tremor (nuisance noise) while preserving the intentional insertion motion. The intent factor α successfully identified the transition to the "Manifold Convergence" phase, effectively freezing the null-space DoFs and preventing the "overshoot-oscillation" cycle common in novices.

TABLE I
TELEOPERATION PERFORMANCE COMPARISON (USB INSERTION)

Method	Success Rate $\uparrow$	Time (s) $\downarrow$	Jerk (10^3) $\downarrow$
Direct Raw	45%	24.5 ± 8.2	15.4
1 Euro Filter	68%	18.2 ± 5.1	8.2
AnyTeleop [44]	72%	16.5 ± 4.3	6.8
VIST (Ours)	95%	11.2 ± 2.1	2.4

C. Evaluation II: Downstream Learning Efficiency

To verify the "Physical Data Regularization" contribution, we trained a **Diffusion Policy** [32] using datasets collected by different teleoperation methods.

1) *Dataset and Training*: For the USB task, we collected 50 demonstrations using AnyTeleop and 50 using VIST. We trained separate Diffusion Policy models (ResNet-18 vision backbone) on subsets of data (10, 20, 30, 50 demos) to evaluate sample efficiency. All models were trained for 1000 epochs.

2) *Policy Rollout Performance*: The trained policies were evaluated on the real robot (20 trials each). As shown in Fig. ??, the policy trained on **VIST-collected data** reached an 80% success rate with only **20 demonstrations**, whereas the policy trained on AnyTeleop data required 50 demonstrations to reach similar performance. **Why?** Visual analysis of the

datasets reveals that VIST data forms a tight "tube" in the state space during the insertion phase, whereas AnyTeleop data exhibits high variance (a "funnel" shape) due to human corrections. VIST effectively reduces the entropy of the demonstration distribution, mitigating *Causal Confusion* [36] by removing spurious correlations between hand jitters and task success.

D. Ablation Study

We validate the necessity of each component in our Intent Modeling (Section III-B). We compared the full VIST against three variants:

- **w/o Geometry** ($\alpha_{geo} = 0$): Intent relies only on velocity. The robot fails to stiffen near the hole, leading to drift.
- **w/o Velocity** ($\alpha_{vel} = 0$): Intent relies only on position. The system feels "sticky" even when the user wants to reposition quickly within the tolerance zone.
- **w/o Direction** ($\alpha_{dir} = 1$): No active gating. The robot fights the user during withdrawal, creating a "trap" effect.

Results indicate that the **Directional Gating** is critical for safety, preventing 90% of the "force fight" incidents during abort maneuvers.

V. DISCUSSION AND LIMITATIONS

Discussion: VIST bridges the gap between the flexibility of vision-based teleoperation and the precision of haptic devices. By shifting the burden of precision from the human's motor cortex to the robot's mathematical observer, we achieve what we term "Cognitive Haptics"—visual cues and stability that substitute for physical force feedback.

Limitations: Currently, VIST requires a prior definition of the task manifold geometry (e.g., the hole axis). While valid for structured assembly, extending this to unstructured tasks (e.g., dough manipulation) remains challenging. Future work will explore learning the task manifold Σ_{task} online using few-shot demonstration learning.

VI. CONCLUSION

In this work, we presented VIST, an intent-aware state estimation framework for low-cost teleoperation. By modeling human intent as a continuous belief factor on $SE(3)$ and dynamically scheduling the covariance of an Adaptive Kalman Filter, VIST solves the "dynamic discontinuity" problem inherent in traditional shared control. Our experiments demonstrate that VIST not only enhances teleoperation performance but also serves as a high-quality data generator, significantly boosting the sample efficiency of downstream Diffusion Policies.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

In this work, we presented VIST (VIST), a probabilistic control framework designed to bridge the "geometric-semantic gap" in low-cost vision-based teleoperation. By reformulating the precision assembly task as an intent-driven

state estimation problem on $SE(3)$, we addressed the fundamental conflict between the need for free-space exploration and manifold-constrained manipulation.

Our contributions are three-fold. First, we derived a continuous **Intent Factor** α based on the principle of least action, serving as a semantic interpreter that translates ambiguous human motions into explicit control confidence. Second, we proposed an **Adaptive Kalman Filter** with null-space covariance scheduling, which mathematically converts the traditional "hard handoff" of shared control into a C^1 -continuous "soft landing," eliminating dynamic jerks without expensive force sensors. Finally, and perhaps most critically for the robot learning community, we demonstrated that VIST acts as a **Physical Data Regularizer**. By compressing nuisance stochasticity at the source, VIST enables downstream policies (e.g., Diffusion Policy) to learn robust behaviors with significantly higher sample efficiency.

VIST effectively democratizes high-precision data collection, allowing consumer-grade cameras to achieve performance metrics previously reserved for high-end haptic devices.

B. Limitations and Future Work

While VIST demonstrates promising results, several avenues remain for future exploration:

1. Online Manifold Estimation: Currently, VIST assumes the task manifold parameters (e.g., the hole axis) are known a priori or calibrated. This limits applicability in unstructured environments. Future work will integrate **Category-Level Object Pose Estimation** to regress task manifolds Σ_{task} online from RGB-D streams, enabling "one-shot" assembly of novel objects.

2. Generalization to Deformable Objects: Our current Lie theoretic formulation assumes rigid body dynamics. Extending the "manifold adsorption" concept to deformable objects (e.g., cable routing, cloth manipulation) requires defining constraints on high-dimensional configuration spaces, which presents a challenging topological problem.

3. VLM-Driven Priors: The current intent model relies on kinematic heuristics. We plan to incorporate **Vision-Language Models (VLMs)** [4] to infer high-level semantic intent (e.g., "screwing" vs. "pushing") from video context. These semantic tokens can be used to dynamically select the structure of the constraint matrix Σ_{cons} , realizing a truly open-world assistive teleoperation system.

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